

MATCHING, ANGULAR FEEDBACK, AND INITIAL-SET DEPENDENCE IN THE PRECISION COST OF CONTRACTING CONTROL SYSTEMS

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ABSTRACT. We study the minimum number of open-loop controls needed to keep every actual initial state within one Euclidean tolerance of a constraint set at all times of a finite horizon. For smooth, globally invertible planar systems whose initial angle affects subsequent radial contraction, we prove a common-cost threshold $\gamma_* = \beta h_a$, where $h_a = -\sum_i a_i \log a_i$ and $\delta_n = \exp(-\gamma n + o(n))$. At and below this threshold, every fixed positive-area compact initial set has rate γ/β . Above it, one compact set of arbitrarily nearly full area has rate h_a simultaneously for all finite higher exponents and their precision sequences, while the full constraint set has a strictly larger lower rate. An invariant cone for moving feasible graphs and a first-deviation estimate compare arbitrary approximate controls with stopped physical witnesses at a loss of at most $1 + Bn$. In a separate radius-autonomous family, unequal matching powers yield distinct full-set and nearly full-area rates at every sufficiently small positive exponent. This contrast identifies the role of matching without asserting a general classification. A scoped topological obstruction excludes coordinate reduction of genuine feedback instances, and a smooth collapse example shows why matching only near the attracting origin is insufficient.

1. INTRODUCTION

Invariance entropy measures the control information required to keep an initial set inside a controlled invariant set; see Kawan [8]. For a system contracting toward a point, outer invariance complexity can be bounded at every fixed positive tolerance. The cost can nevertheless grow exponentially when the tolerance decreases with the horizon. Which features of feasible control choices determine that growth, and when does it depend on the actual initial set?

We consider all binary inputs and assign a member of a finite control catalogue to each physical initial state. The same tolerance δ_n is imposed at every integer time from 0 through n ; departures from and returns to the constraint set are permitted within that tolerance. Our question concerns independence of the exponential cost across *all* fixed positive-area compact initial sets. A set of almost full area need not contain the rare states that determine the full-set cost, so positive area alone does not settle this question.

The main theorem establishes an exact common-cost boundary for a matched angular-feedback family. Here even a fixed control word on an initial-radius fibre produces different radial trajectories as the initial angle varies. A logarithmic slope cone controls the moving feasible graphs and proves that complete feasible initial sections remain intervals. We then estimate a wrong first branch in physical coordinates. These estimates, together with complete tails, connect classical stopping counts to the optimal catalogue of arbitrary approximate controls. The same nearly full-area compact set works for *every* higher precision exponent and its precision sequences.

A second theorem concerns a different, radius-autonomous family. When its two radial powers differ, nearly full-area sets and the full constraint already have distinct rates at

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arbitrarily small positive precision exponents. This yields a matching criterion within that family, rather than a necessary and sufficient classification of angular-feedback systems. A topological feasible-image obstruction shows that a nonempty subfamily of the main result is not a simultaneous coordinate corollary of the radius-autonomous models. A local-matching collapse example delineates another restriction of the positive theorem.

1.1. Relation to previous work. The finite-horizon object below is the outer invariance complexity of Chen and Zhong [1, Section 2.2], with $n + 1$ constrained times instead of n . Their bounded-complexity and equi-invariability results concern fixed tolerances. Exponential and practical stabilization entropies [3, 4] address accuracy in different stabilization tasks. These works give the information context; we do not introduce a new catalogue definition.

Measure-theoretic invariance entropies and variational principles are developed in [10, 9]. The entropy dimensions of Chen, Huang and Zhong [2] use prescribed invariant-partition cylinders and weights $\exp[-s(N\tau)^\alpha]$; their measure-covering tolerance permits missing measure. Their proofs provide cylinder covering and variational tools, but do not estimate Euclidean first-deviation bands for arbitrary controls at δ_n . On our connected Q , a finite clopen invariant partition would have only one nonempty cell, whereas neither single input preserves all of Q at the first step. Their main clopen variational theorems therefore do not directly apply. This does not dispose of their measurable cylinder estimates: those still require the physical comparison proved here. Ordinary area need not be invariant, and its use here is not a new measure-theoretic framework.

Graph and volume-distortion methods in control entropy are existing tools; Kawan and Da Silva [7, Appendix] prove a graph-transform volume lemma near a partially hyperbolic all-time lift. In our uniformly contracting family, every bounded bi-infinite trajectory is the origin: $\|x_0\|_* \leq L^k \sup_j \|x_j\|_*$ for all k . The physical angular coordinate is singular there. That volume lemma alone does not give the complete feasible-graph splits or a uniform diagonal-precision comparison for all approximate controls.

Prefix stopping, product weights, types, and typical sets are classical [6, 5]. Yang, Chen, Yang and Zhou [11] develop cost-threshold pressure and Bowen-equation results on a given invariant partition. Their prescribed-address spanning/separated comparison is not a comparison with all approximate physical inputs. The additional controlled step here is a first-deviation band and a stopped-witness count with only linear loss. The mismatch root and the type estimates are applications of existing counting tools, not new pressure or graph-transform theory.

2. COUNTING PROBLEM AND MAIN RESULTS

Let

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For any two maps $F_0, F_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and $u = (u_1, u_2, \dots) \in \{0, 1\}^{\mathbb{N}}$, write $F_u^0 x = x$ and $F_u^k x = F_{u_k} \circ \dots \circ F_{u_1} x$. Distances are Euclidean, and $Q^\delta = \{x : \text{dist}(x, Q) < \delta\}$.

Definition 2.1. For a nonempty compact $K \subset Q$, let $r_F(n, \delta; K)$ be the least size of a finite catalogue $\mathcal{U} \subset \{0, 1\}^{\mathbb{N}}$ such that

$$(2.1) \quad \forall x \in K \ \exists u \in \mathcal{U} \ \forall k \in \{0, \dots, n\} : \quad F_u^k x \in Q^\delta.$$

The input assignment may depend on the actual state. No exact itinerary is imposed. A trajectory may leave and reenter Q provided every required time satisfies (2.1). Only the first n letters matter, but infinite inputs make the tail constructions unambiguous. Finiteness will follow from exact controlled feasibility.

Throughout, a precision sequence is positive, tends to zero, and satisfies $-\log \delta_n/n \rightarrow \gamma$ for a finite $\gamma \geq 0$. It need not be monotone. All logarithms are natural. Fix

$$(2.2) \quad a_0, a_1 \in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta > 1,$$

and put $a_* = \min(a_0, a_1)$ and $h = -a_0 \log a_0 - a_1 \log a_1$. We also use

$$H_b(t) = -t \log t - (1-t) \log(1-t), \quad \chi_a(t) = -t \log a_0 - (1-t) \log a_1,$$

with $0 \log 0 = 0$, so $H_b(a_0) = \chi_a(a_0) = h$.

2.1. Matched angular feedback. For

$$(2.3) \quad 0 \leq \varepsilon < \frac{a_*}{16(\beta+1)},$$

define on the whole plane

$$(2.4) \quad p_0(s, z) = a_0 + \frac{\varepsilon z}{1+s^2+z^2}, \quad p_1(s, z) = a_1 - \frac{\varepsilon z}{1+s^2+z^2},$$

$$(2.5) \quad H_0(s, z) = (sp_0^\beta, p_0^{\beta-1}(z + p_1 s)), \quad H_1(s, z) = (sp_1^\beta, p_1^{\beta-1}(z - p_0 s)).$$

All factors in (2.5) are evaluated at the input state. The endpoint $\varepsilon = 0$ is the matched linear family.

Theorem 2.2 (Common-cost boundary with angular feedback). *The maps (2.5) are global C^∞ diffeomorphisms, fix the origin, drive bounded initial sets uniformly exponentially to the origin under all inputs, and make Q controlled invariant. Their derivatives at the origin are those of the matched linear maps. For every fixed positive-area compact $K \subset Q$ and every precision sequence of finite exponent $\gamma \geq 0$,*

$$(2.6) \quad \min\{h, \gamma/\beta\} \leq \liminf_n \frac{\log r_H(n, \delta_n; K)}{n} \leq \limsup_n \frac{\log r_H(n, \delta_n; K)}{n} \leq \min\{\log 2, \gamma/\beta\}.$$

In particular, for $0 \leq \gamma \leq \beta h$ the limit exists and equals γ/β .

For every $\zeta \in (0, 1)$ there is a fixed compact $K_\zeta \subset Q$ of area greater than $1 - \zeta$ such that, simultaneously for every finite $\gamma > \beta h$ and every precision sequence with that exponent,

$$(2.7) \quad \lim_n \frac{\log r_H(n, \delta_n; K_\zeta)}{n} = h < \liminf_n \frac{\log r_H(n, \delta_n; Q)}{n}.$$

The set is chosen before the exponent, sequence, and horizon.

The threshold concerns all positive-area compact initial sets. It does not classify them into two possible costs above the threshold, nor determine the exact high-precision spectrum or its limit for Q . The positive result holds on the entire range (2.2)–(2.3); the coordinate obstruction in Section 7 has a smaller scope.

2.2. A different family: radius-autonomous mismatch. Keep a_0, a_1 as above, but now let $\beta_0, \beta_1 > 1$ and $\beta_{\max} = \max_i \beta_i$. For

$$(2.8) \quad 0 \leq \varepsilon < \frac{a_*}{4(\beta_{\max} + 1)}, \quad q_0(s) = a_0 + \frac{\varepsilon s}{1+s^2}, \quad q_1(s) = a_1 - \frac{\varepsilon s}{1+s^2},$$

define

$$(2.9) \quad \begin{aligned} G_0(s, z) &= (sq_0(s)^{\beta_0}, q_0(s)^{\beta_0-1}(z + q_1(s)s)), \\ G_1(s, z) &= (sq_1(s)^{\beta_1}, q_1(s)^{\beta_1-1}(z - q_0(s)s)). \end{aligned}$$

Set

$$(2.10) \quad \rho_i = a_i^{\beta_i}, \quad \chi = -\sum_i a_i \log \rho_i, \quad \gamma_0 = -\log \max_i \rho_i, \quad \rho_0^D + \rho_1^D = 1 \quad (D > 0).$$

Theorem 2.3 (Low-precision separation under radial mismatch). *Suppose $\beta_0 \neq \beta_1$. The maps (2.9) are global smooth diffeomorphisms with a common attracting origin under all inputs, and Q is controlled invariant. For every $\zeta \in (0, 1)$ there is one compact $K_\zeta \subset Q$ of area greater than $1 - \zeta$ such that, simultaneously for every $0 < \gamma < \gamma_0$ and every precision sequence of exponent γ ,*

$$(2.11) \quad \lim_n \frac{\log r_G(n, \delta_n; Q)}{n} = \gamma D, \quad \lim_n \frac{\log r_G(n, \delta_n; K_\zeta)}{n} = \gamma h/\chi, \quad D > h/\chi.$$

Every fixed positive-area compact $K \subset Q$ also has lower rate at least $\gamma h/\chi$ in this range.

Thus a fixed system can exhibit strict initial-set dependence at every sufficiently small positive exponent. The theorem is restricted to (2.9); it makes no claim about unequal powers in the angular-feedback family. The precise matching criterion in this radius-autonomous class is given in Corollary 6.4.

3. MOVING FEASIBLE GRAPHS AND PHYSICAL DISTORTION

In this section the system is (2.5). Put

$$p_- = a_* - \varepsilon/2, \quad p_+ = 1 - p_-, \quad r = p_+^\beta < 1, \quad \sigma_0 = -1, \quad \sigma_1 = 1.$$

These bounds hold on the whole plane, not just on Q .

Lemma 3.1 (Global dynamics and one-step feasibility). *The dynamical assertions of Theorem 2.2 hold. For $s > 0$, write $y = z/s$ and $P_i(s, y) = p_i(s, sy)$. There is a unique $\theta(s) \in (-1, 1)$ with*

$$(3.1) \quad \theta(s) = 2P_0(s, \theta(s)) - 1.$$

The complete feasible angular domains are $[-1, \theta(s)]$ for input 0 and $[\theta(s), 1]$ for input 1. Choosing 0 at the shared endpoint defines a natural exactly feasible code.

Proof. For $f = z/(1 + s^2 + z^2)$, direct differentiation gives

$$(3.2) \quad |f| \leq \frac{1}{2}, \quad |sf_s| \leq 1, \quad |zf_z| \leq \frac{1}{2}, \quad |sf_z| \leq \frac{1}{2}.$$

Use $f_s = -2sz/(1 + s^2 + z^2)^2$, $f_z = (1 + s^2 - z^2)/(1 + s^2 + z^2)^2$, and $2|sz| \leq s^2 + z^2$ to obtain these bounds. In the shear $v = z - \sigma_i s$, with output shear $V = Z - \sigma_i S$, the map is $(S, V) = (sp_i^\beta, vp_i^{\beta-1})$. Consequently

$$\det DH_i = p_i^{2\beta-2} [p_i + \beta s(p_i)_s + ((\beta - 1)z + \sigma_i s)(p_i)_z] > 0,$$

since the added terms have absolute value at most $3\beta\varepsilon/2 < p_-$. To prove global bijectivity, for any output (S, V) solve

$$t - p_i(St^{-\beta}, Vt^{1-\beta} + \sigma_i St^{-\beta}) = 0, \quad p_- \leq t \leq p_+.$$

The left side has opposite weak signs at the endpoints. Its derivative is at least $1 - 3\beta\varepsilon/(2p_-) > 0$, by (3.2). It has exactly one root; that root recovers the unique preimage. The implicit function theorem supplies the smooth global inverse. Positive factors make the real powers smooth, and differentiation at zero gives

$$DH_0(0) = \begin{pmatrix} a_0^\beta & 0 \\ a_0^{\beta-1}a_1 & a_0^{\beta-1} \end{pmatrix}, \quad DH_1(0) = \begin{pmatrix} a_1^\beta & 0 \\ -a_1^{\beta-1}a_0 & a_1^{\beta-1} \end{pmatrix}.$$

Choose $M \geq 1$ with $M > r/(1 - p_+^{\beta-1})$, and set

$$(3.3) \quad \|x\|_* = \max(M|s|, |z|), \quad L = \max\{r, p_+^{\beta-1} + r/M\} < 1, \quad \|H_i x\|_* \leq L\|x\|_*.$$

Indeed $|s'| \leq r|s|$ and $|z'| \leq p_+^{\beta-1}|z| + r|s|$ on the whole plane. This proves convergence relative to the origin, without asserting pairwise contraction, and controls tails outside Q .

In calculation coordinates the actual map is

$$(3.4) \quad S = sP_i^\beta, \quad Y = \sigma_i + (y - \sigma_i)/P_i.$$

On $0 < s \leq 1$, $|y| \leq 1$,

$$(3.5) \quad |(P_i)_s| \leq \varepsilon, \quad |(P_i)_y| \leq \varepsilon s.$$

The separating function $y - 2P_0(s, y) + 1$ has positive derivative at least $1 - 2\varepsilon$ and opposite signs at $-1, 1$. Its unique zero is (3.1). The condition $-1 \leq Y \leq 1$ is exactly the asserted pair of domains; their angular endpoints map to $-1, 1$. Since $S < s \leq 1$, these are the full one-step constraints. Both controls fix the vertex, completing controlled invariance. $\square$

Lemma 3.2 (Invariant graph cone and complete intervals). *Fix an actual initial radius $0 < s_0 \leq 1$. For every finite word w , the set $I_w(s_0)$ of initial angles whose trajectory remains in Q at all times through $|w|$ is a nondegenerate closed interval. Its terminal angular map is increasing onto $[-1, 1]$. The terminal radial graph $s = S(y)$ over that entire interval satisfies*

$$(3.6) \quad |S'(y)/S(y)| \leq 1.$$

Every complete prefix-free family gives a cover of $[-1, 1]$ by these intervals, with disjoint interiors and shared endpoints retained.

Proof. Suppose a feasible-prefix image is a graph obeying (3.6). Write $k = S'/S$ and, along that graph, $A = (P_i)_s S' + (P_i)_y$, so $|A| \leq 2\varepsilon S$. By (3.4),

$$(3.7) \quad q_i = 1 - (y - \sigma_i)A/P_i, \quad \frac{dY}{dy} = q_i/P_i, \quad k_{\text{new}} = (P_i k + \beta A)/q_i.$$

The parameter bound gives $p_- > 63a_*/64$ and $(2\beta + 4)\varepsilon < 3a_*/16 < p_-/2$. Since $P_i(1 - P_i) \geq p_-(1 - p_-) \geq p_-/2$, we have

$$(3.8) \quad q_i \geq 1 - 4\varepsilon S/P_i > 0, \quad |k_{\text{new}}| \leq \frac{P_i + 2\beta\varepsilon S}{1 - 4\varepsilon S/P_i} < 1.$$

Along the graph, $\Psi(y) = y - 2P_0(S(y), y) + 1$ has derivative at least $1 - 4\varepsilon S > 0$ and opposite endpoint signs. It therefore divides the parent graph at exactly one point. On each feasible side, (3.4) is increasing onto the entire $[-1, 1]$, and (3.8) preserves the cone. The original fibre has $S \equiv s_0$, hence slope zero. Induction proves all assertions, retaining every intermediate constraint and closed endpoint. The terminal angular map has positive derivative, including one-sided endpoint derivatives. Smooth dependence of the endpoints on $s_0 > 0$ follows from the implicit function theorem. $\square$

For $w = w_1 \cdots w_\ell$, put $\lambda_w = \prod_{j=1}^\ell a_{w_j}$ and $\mathcal{P}_w(x) = \prod_{j=1}^\ell p_{w_j}(x_{j-1})$. The actual product depends on the initial angle; there is no shared radial trajectory on an initial-radius fibre. Write

$$A_w = \{(s_0, s_0 y) : 0 < s_0 \leq 1, y \in I_w(s_0)\} \cup \{(0, 0)\}$$

for its physical complete feasible initial strip.

Lemma 3.3 (Derived radial and angular comparisons). *There are fixed $C_P, C \geq 1$ such that, for every complete feasible word and every point of its initial interval,*

$$(3.9) \quad C_P^{-1} \lambda_w \leq \mathcal{P}_w(x) \leq C_P \lambda_w, \quad C_P^{-\beta} s_0 \lambda_w^\beta \leq s_\ell \leq C_P^\beta s_0 \lambda_w^\beta, \quad C^{-1} \lambda_w^{-1} \leq \partial_{y_0} Y_{w, s_0} \leq C \lambda_w^{-1}.$$

Consequently

$$(3.10) \quad 2C^{-1} \lambda_w \leq |I_w(s_0)| \leq 2C \lambda_w,$$

and the physical complete feasible strip A_w satisfies

$$(3.11) \quad C^{-1} \lambda_w \leq \text{area}(A_w) \leq C \lambda_w.$$

All finite-word boundary curves have zero area and remain in the covers.

Proof. Along a feasible word, $s_j \leq r^j s_0$ and $|p_i(x_j) - a_i| \leq \varepsilon s_j$. Thus

$$|\log(\mathcal{P}_w/\lambda_w)| \leq B_P := \frac{\varepsilon}{p_-(1-r)}, \quad C_P = e^{B_P}.$$

Set $d_0 = 4\varepsilon/p_- < 1$. Equation (3.7) gives $|q_i - 1| \leq 4\varepsilon s_j/p_-$, and hence

$$\sum_j |\log q_i| \leq B_q := \frac{4\varepsilon}{p_-(1-d_0)(1-r)}.$$

The chain rule along the moving graphs yields $\partial_{y_0} Y_{w,s_0} = \prod_j q_{w_{j+1}}/p_{w_{j+1}}$. Taking $C = e^{B_P+B_q}$ proves the angular comparison. The radial identity $s_\ell = s_0 \mathcal{P}_w^\beta$ proves the remaining inequalities in (3.9). The full-range assertion of Lemma 3.2 gives (3.10). Integrating the length against the physical Jacobian $s_0 ds_0$ proves (3.11), since $\int_0^1 s_0 ds_0 = 1/2$. Smooth endpoints for $s_0 > 0$ and Fubini give zero area to each boundary curve. Their countable union, and the fixed vertex, are null for the subsequent area argument. $\square$

4. ARBITRARY APPROXIMATE INPUTS AND STOPPED PHYSICAL WITNESSES

For any input u , denote its actual admissible initial subset by $V(n, \delta, u) = \{x \in Q : H_u^j x \in Q^\delta, 0 \leq j \leq n\}$. The natural code is a proof tool; it does not restrict these inputs.

Lemma 4.1 (Physical first-deviation band). *If u covers $(s_0, s_0 y_0)$ and first differs from its natural exact code at time j , let $a = u|_{j-1}$. There is a separating point $b_a(s_0) \in I_a(s_0)$, depending only on that prefix and radius, such that*

$$(4.1) \quad |y_0 - b_a(s_0)| \leq C_b \delta \lambda_a^{1-\beta}/s_0, \quad C_b = \frac{2C_P^\beta p_-^{1-\beta} C}{1-4\varepsilon}.$$

For fixed u, j the possible points lie on one side of this moving boundary and have physical area at most $C_b \delta \lambda_a^{1-\beta}$.

Proof. The common prefix is exact, and its image is the full graph of Lemma 3.2. Let Y_* be its unique separating zero of $\Psi(Y) = Y - 2P_0(S(Y), Y) + 1$, and let $b_a = Y_{a,s_0}^{-1}(Y_*)$. If the differing branch is i , direct substitution gives

$$(4.2) \quad |Y_j| - 1 = |\Psi(Y)|/p_i, \quad |z_j| - s_j = S p_i^{\beta-1} |\Psi(Y)|.$$

At the shared endpoint both excesses are zero; either input is included. A necessary condition for Q^δ is $|z_j| - s_j < 2\delta$, by comparison with a point of Q at Euclidean distance less than δ . Use $S \geq C_P^{-\beta} s_0 \lambda_a^\beta$ and $p_i \geq p_-$ in (4.2). The derivative of $\Psi(Y_{a,s_0}(y_0))$ is at least $(1-4\varepsilon)C^{-1} \lambda_a^{-1}$. The mean value theorem proves (4.1). The branch u_j fixes its side. Integrating the one-sided width against $s_0 ds_0$ cancels $1/s_0$ and gives the asserted physical area, without a lower radial cutoff. Later reentry does not remove the constraint at time j . No cone estimate is imposed on the subsequent approximate trajectory. $\square$

Let $\mathcal{L}(n, t)$ stop every branch at its first word with $\lambda_w \leq t$, or at depth n if no earlier crossing occurs. Write $N(n, t) = \#\mathcal{L}(n, t)$, for $0 < t < 1$. This is a complete prefix-free counting tree, not an assumed optimal catalogue. It satisfies

$$(4.3) \quad \sum_{w \in \mathcal{L}(n, t)} \lambda_w = 1, \quad \lambda_w > a_* t, \quad N(n, t) \leq (a_* t)^{-1}.$$

These follow by replacing a parent weight by its two child weights, and observing that every proper prefix of a leaf precedes a crossing.

Lemma 4.2 (Comparison with the actual catalogue). *For fixed constants B, C_0, C_U , all $n \geq 1$ and all sufficiently small $\delta > 0$,*

$$(4.4) \quad \frac{N(n, \delta^{1/\beta})}{1 + Bn} \leq r_H(n, \delta; Q) \leq N(n, (\delta/C_0)^{1/\beta}) \leq C_U \delta^{-1/\beta}.$$

Also $r_H(n, \delta; Q) \leq 2^n$.

Proof. For the lower bound take $t = \delta^{1/\beta}$ and choose the physical witness $(1, y_w)$ at the midpoint of each $I_w(1)$. The intervals partition the fibre. By (3.10) and (4.3), ordered witnesses have successive gaps greater than $2C^{-1}a_*t$. An arbitrary input u has at most one stopping leaf as a prefix. Every other covered witness first differs at some $j \leq n$, with the one common prefix $a = u|_{j-1}$ and $\lambda_a > t$. At radius 1, (4.1) confines these witnesses to one side of a fixed centre in a band of width at most $C_b \delta t^{1-\beta} = C_b t$. Spacing limits their number to $B = \lceil 1 + CC_b/(2a_*) \rceil$. Thus u covers at most $1 + Bn$ witnesses. Their midpoints are interior to every ancestor, so their natural prefixes are precisely the chosen leaves. Any catalogue covering Q must cover all witnesses, proving the lower bound.

For the upper bound take $t = (\delta/C_0)^{1/\beta}$, use every leaf as an exact prefix, and append the same infinite tail, say 0^∞ . Lemma 3.2 covers every initial fibre with all intermediate constraints. A depth- n leaf needs no further constrained time. An earlier stopped leaf has radius at most $C_P^\beta t^\beta$, so its terminal adapted norm is at most $MC_P^\beta t^\beta$. With $C_0 = 2\sqrt{2}MC_P^\beta$, (3.3) bounds the Euclidean norm of its entire remaining tail by $\delta/2$. This is strictly inside Q^δ at every later required time. The apex is fixed, and all closed endpoints have been retained. Equation (4.3) gives $C_U = C_0^{1/\beta}/a_*$. Alternatively every exactly feasible length- n prefix, extended arbitrarily, gives the 2^n upper bound. $\square$

5. AREA-TYPICAL INITIAL STATES AND PROOF OF THE THRESHOLD

Let $c_k(x)$ denote the first k letters of the natural exact code, with the vertex assigned 0^∞ . The continuous maps and separating curves make every finite assigned-code set Borel measurable.

Lemma 5.1 (Physical typical sets). *For area-almost every $x \in Q$, the frequency of zero symbols in $c_k(x)$ tends to a_0 . Every compact $K \subset Q$ of positive area contains a compact positive-area set T on which this convergence is uniform. For every $0 < \eta < h$, there is $D_\eta \geq 1$ such that*

$$(5.1) \quad D_\eta^{-1} e^{-(h+\eta)k} \leq \lambda_{c_k(x)} \leq D_\eta e^{-(h-\eta)k} \quad (x \in T, k \geq 0).$$

For each $\zeta \in (0, 1)$ such a compact T can be chosen with $\text{area}(T) > 1 - \zeta$ inside Q .

Proof. Normalize area on Q , whose area is one. For every assigned word w , (3.11) bounds its probability by $C\lambda_w$. Thus the area of states whose length- k prefix has zero frequency at distance at least η from a_0 is at most

$$(5.2) \quad C(k+1)e^{-2\eta^2 k}.$$

Indeed, for $t = j/k$, put $D_{\text{KL}}(t||a_0) = t \log(t/a_0) + (1-t) \log((1-t)/a_1)$. Then

$$\binom{k}{j} a_0^j a_1^{k-j} \leq e^{-k D_{\text{KL}}(t||a_0)}, \quad D_{\text{KL}}(t||a_0) \geq 2(t - a_0)^2.$$

The first inequality uses $\binom{k}{j} \leq e^{k H_b(j/k)}$: the binomial mass at j with parameter j/k is at most one. The endpoint types follow by continuity. The second inequality follows from $D_{\text{KL}}''(t) = 1/[t(1-t)] \geq 4$, with its value and first derivative zero at a_0 . Summing over types gives (5.2). Borel–Cantelli, followed by a countable sequence of positive η tending to zero, proves almost everywhere convergence. No independence of the state-dependent digits has been used.

Remove the null set of finite-word boundary curves and the vertex. Egorov's theorem and inner regularity of area give a compact subset of any positive-area K with positive area and uniform frequency convergence. On Q they give a compact set of area greater than $1 - \zeta$. Since $-\log \lambda_{c_k(x)}/k$ is χ_a of the zero frequency, it converges uniformly to $\chi_a(a_0) = h$ on each chosen set. Absorb the finitely many early times into D_η to obtain (5.1). All sets are chosen before any precision exponent or sequence. $\square$

Lemma 5.2 (Area covered by one arbitrary input). *If T satisfies (5.1), then for each $0 < \eta < h$ there is a constant C_η such that, uniformly in u, n, δ and $0 \leq m \leq n$,*

$$(5.3) \quad \text{area}(T \cap V(n, \delta, u)) \leq C_\eta \left[e^{-(h-\eta)m} + \delta e^{(\beta-1)(h+\eta)m} \right].$$

Proof. Agreement through m confines the state to $A_{u|m}$. If this agreement class meets T , its reference product is at most $D_\eta e^{-(h-\eta)m}$, and (3.11) gives the first term.

For a first disagreement at $j \leq m$, a nonempty class in T has common prefix $a = u|_{j-1} = c_{j-1}(x)$ for some $x \in T$. Equation (5.1) bounds its product from below by $D_\eta^{-1} e^{-(h+\eta)(j-1)}$. Lemma 4.1 bounds its area by $C_b D_\eta^{\beta-1} \delta e^{(\beta-1)(h+\eta)(j-1)}$. The sum over $j \leq m$ is a geometric sum with ratio greater than one, proving the second term. Empty classes give no contribution. Later disagreement times are already included in the agreement class through m . $\square$

Proof of Theorem 2.2. The dynamical assertions follow from Lemma 3.1. For a given positive-area compact K , choose $T \subset K$ as in Lemma 5.1. For small δ set

$$m = \min \left\{ n, \left\lfloor \frac{-\log \delta}{\beta(h+\eta)} \right\rfloor \right\}.$$

Then $\delta e^{\beta(h+\eta)m} \leq 1$, and Lemma 5.2 gives

$$\text{area}(T \cap V(n, \delta, u)) \leq 2C_\eta e^{-(h-\eta)m}.$$

Every catalogue for K must cover T , so

$$(5.4) \quad r_H(n, \delta; K) \geq \frac{\text{area}(T)}{2C_\eta} e^{(h-\eta)m}.$$

For $-\log \delta_n/n \rightarrow \gamma$, passage to rates and then $\eta \downarrow 0$ yields the lower bound $\min\{h, \gamma/\beta\}$. Lemma 4.2 and $K \subset Q$ give both upper bounds in (2.6). They coincide with the lower bound when $\gamma \leq \beta h$. The argument includes $\gamma = 0$ and equality at the threshold, and needs no monotonicity of δ_n .

For the high side choose once a compact T_ζ from Lemma 5.1 with area greater than $1 - \zeta$, and set $K_\zeta = T_\zeta \cup \{(0, 0)\}$. For every $\eta > 0$ and all sufficiently large n , all natural prefixes on T_ζ have zero frequency within η of a_0 . The number of these prefixes is at most

$$(5.5) \quad (n+1) \exp \left[n \max_{\substack{0 \leq t \leq 1 \\ |t-a_0| \leq \eta}} H_b(t) \right].$$

Use each as an exact length- n prefix with any infinite extension, and one extra input for the vertex. This covers K_ζ exactly for all required times, for every $\delta > 0$. Continuity of H_b gives upper rate h . For every finite $\gamma > \beta h$, (2.6) gives lower rate h on this same set. Its rate therefore exists and equals h simultaneously for all those exponents and all their precision sequences.

It remains to obtain a strictly larger lower rate on Q . Since $a_0 \neq a_1$,

$$\Delta = \chi_a(1/2) - h = (a_0 - 1/2) \log(a_0/a_1) > 0.$$

For a given $\gamma > \beta h$, let

$$(5.6) \quad \theta_\gamma = \min \left\{ \frac{1}{2}, \frac{\gamma/\beta - h}{2\Delta} \right\}, \quad t_\gamma = (1 - \theta_\gamma)a_0 + \theta_\gamma/2.$$

By affinity of χ_a and strict concavity of H_b , $\chi_a(t_\gamma) < \gamma/\beta$ and $H_b(t_\gamma) > h$. Choose integers j_n with $j_n/n \rightarrow t_\gamma$. For all sufficiently large n , every word of length n with exactly j_n zeros has

$$-\log \lambda_w = n\chi_a(j_n/n) < -\log \delta_n/\beta.$$

Every proper prefix has still smaller cost. All these words are therefore terminal leaves of $\mathcal{L}(n, \delta_n^{1/\beta})$. The elementary type bound $\binom{n}{j} \geq e^{nH_b(j/n)}/(n+1)$ and Lemma 4.2 imply

$$(5.7) \quad r_H(n, \delta_n; Q) \geq \frac{\exp[nH_b(j_n/n)]}{(n+1)(1+Bn)}.$$

For completeness, the type bound follows since j is a mode of the binomial law with parameter j/n , so its mass is at least $1/(n+1)$; endpoint types are immediate. Taking lower rates in (5.7) yields $H_b(t_\gamma) > h$, proving (2.7). The strict gap may tend to zero as $\gamma \downarrow \beta h$. $\square$

Corollary 5.3 (Matched linear systems). *For*

$$A_0(s, z) = (a_0^\beta s, a_0^{\beta-1}(z + a_1 s)), \quad A_1(s, z) = (a_1^\beta s, a_1^{\beta-1}(z - a_0 s)),$$

the conclusions of Theorem 2.2 hold with the same threshold βh .

Proof. Set $\varepsilon = 0$ in the theorem. Then $\mathcal{P}_w(x) = \lambda_w$ and the product and angular comparison constants are $C_P = C = 1$. $\square$

6. RADIUS-AUTONOMOUS MATCHING AND MISMATCH

We now use (2.9), with parameters (2.8). The quantities in this section are independent of the feedback constants. For words let $\lambda_w = \prod_j a_{w_j}$ and $R_w = \prod_j \rho_{w_j}$. Write $\beta_{\min} = \min_i \beta_i$ and put

$$q_- = a_* - \varepsilon/2, \quad q_+ = 1 - q_-, \quad r_R = q_+^{\beta_{\min}} < 1, \\ C_R^P = \exp\left(\frac{\varepsilon}{q_-(1-r_R)}\right), \quad C_R^s = (C_R^P)^{\beta_{\max}}.$$

Lemma 6.1 (Actual strips and radial products). *The maps (2.9) are global smooth diffeomorphisms and admit a common origin-contracting norm on the whole plane. At a fixed initial radius, each complete feasible word has a closed angular interval $I_w(s)$, and every complete prefix-free tree partitions that fibre with endpoints included. Along a word set*

$$\mathcal{Q}_w(s) = \prod_j q_{w_{j+1}}(s_j), \quad \mathcal{R}_w(s) = \prod_j q_{w_{j+1}}(s_j)^{\beta_{w_{j+1}}}.$$

Then, uniformly in word and radius $0 \leq s \leq 1$,

$$(6.1) \quad (C_R^P)^{-1} \lambda_w \leq \mathcal{Q}_w \leq C_R^P \lambda_w, \quad (C_R^s)^{-1} R_w \leq \mathcal{R}_w \leq C_R^s R_w,$$

$s|_w = s\mathcal{R}_w(s)$ and $|I_w(s)| = 2\mathcal{Q}_w(s)$. Physical strip area is between $(C_R^P)^{-1} \lambda_w$ and $C_R^P \lambda_w$.

Proof. For $f(s) = s/(1+s^2)$, $|f| \leq 1/2$ and $|sf'| \leq 1/2$. Thus $g_i(s) = sq_i(s)^{\beta_i}$ has derivative

$$g'_i(s) = q_i^{\beta_i-1}(q_i + \beta_i s q'_i) > 0,$$

with the bracket at least $a_* - (\beta_{\max} + 1)\varepsilon/2 > 0$. Since $q_i(s) \rightarrow a_i$ at infinity, g_i is onto $\mathbb{R}$. Its smooth inverse recovers s , then the positive coefficient $q_i^{\beta_i-1}$ recovers z . This proves global invertibility. With $M_R \geq 1$, $M_R > r_R/(1 - q_+^{\beta_{\min}-1})$, the norm $\max(M_R|s|, |z|)$ contracts relative to the origin by at most $\max\{r_R, q_+^{\beta_{\min}-1} + r_R/M_R\} < 1$.

At radius s , the angular maps are $T_{0,s}(y) = (y + q_1)/q_0$ and $T_{1,s}(y) = (y - q_0)/q_1$. The full domains are $[-1, \xi(s)]$, $[\xi(s), 1]$, where $\xi = q_0 - q_1$, with inverse branches $B_{0,s}(y) = q_0(s)y - q_1(s)$ and $B_{1,s}(y) = q_1(s)y + q_0(s)$. The radius after the first letter does not depend on the angle, so

$$(6.2) \quad I_{iv}(s) = B_{i,s}(I_v(g_i(s))).$$

This induction retains all intermediate constraints and gives the partition and length assertions. Controlled feasibility follows.

The radial path obeys $s_j \leq r_R^j s$. Since $|q_i(s_j) - a_i| \leq \varepsilon s_j$, summing logarithmic errors proves (6.1); the radial error is at most $\beta_{\max}$ times the angular one. Integration against $s ds$ gives the area bounds. Finite-word boundary curves have zero area, and the vertex is fixed. $\square$

Lemma 6.2 (Relative radial-stop witness comparison). *Let $\mathcal{L}_R(n, t)$ stop words at first $R_w \leq t$, or at depth n , and let $N_R(n, t)$ be its size. Fixed constants $B_R, C_{0,R}$ satisfy*

$$(6.3) \quad \frac{N_R(n, \delta)}{1 + B_R n} \leq r_G(n, \delta; Q) \leq N_R(n, \delta/C_{0,R})$$

for all horizons and sufficiently small positive tolerances. For a first difference at time j , common prefix a , an arbitrary covered physical state lies in a one-sided band of width

$$(6.4) \quad |y - b_a(s)| \leq C_{b,R} \delta \lambda_a / (s R_a), \quad C_{b,R} = 2C_R^P C_R^s \max_i (a_i / \rho_i).$$

Its physical area is at most $C_{b,R} \delta \lambda_a / R_a$.

Proof. The common word is exact. Its angle map is affine with slope $Q_a(s)^{-1}$. For the differing branch i , the excess at the first difference is

$$|y_j| - 1 = |y - b_a(s)| / Q_{ai}(s), \quad s_j = s \mathcal{R}_{ai}(s).$$

The necessary condition $|z_j| - s_j < 2\delta$ and (6.1) give (6.4). The side is fixed by the differing input; integration cancels $1/s$. This includes ties and subsequent reentry.

For the lower bound use midpoint witnesses at radius 1 for leaves stopped at $t = \delta$. Write $\rho_* = \min_i \rho_i$. Every stopped or capped leaf satisfies $R_w > \rho_* t$. For any descendant $w = av$ of a fixed proper prefix a , $\lambda_v \geq R_v = R_w / R_a > \rho_* t / R_a$, because $\beta_i > 1$. By Lemma 6.1, all leaf intervals below a have length greater than $2(C_R^P)^{-1} \lambda_a \rho_* t / R_a$. Their ordered midpoints have at least that spacing. For one input and one first-difference time, (6.4) at $s = 1$ confines these descendant witnesses to a band of width $C_{b,R} \delta \lambda_a / R_a$. Hence at most

$$B_R = \left\lceil 1 + \frac{C_{b,R} C_R^P}{2\rho_*} \right\rceil$$

can be covered at that time. Add the one possible leaf prefix of the input and sum over times to get $1 + B_R n$. This is a conditional spacing comparison within the shared prefix, rather than a global minimum-length assertion.

For the upper bound, the full leaf intervals of (6.2) cover each physical fibre exactly through the leaf. Append the same infinite tail to each leaf at $t = \delta / C_{0,R}$, with $C_{0,R} = 2\sqrt{2} M_R C_R^s$. An early terminal radius is at most $C_R^s t$, and the global origin-contraction norm of Lemma 6.1 keeps the entire tail within Euclidean distance $\delta/2$ of the origin. Capped leaves are exactly feasible through n . All endpoints and the vertex are covered, proving (6.3). $\square$

Proof of Theorem 2.3. Lemma 6.1 proves the dynamical assertions. Write $V_G(n, \delta, u) = \{x \in Q : G_u^j x \in Q^\delta, 0 \leq j \leq n\}$ for the actual all-time initial set. For $0 < \gamma < \gamma_0$, the bound $R_w \leq e^{-\gamma_0 n}$ at depth n implies that every branch crosses either of the thresholds in (6.3) by time n , eventually for any precision sequence of exponent γ . For an uncapped

radial stopping tree, replacement of a parent weight R_w^D by its children preserves total weight, since $\rho_0^D + \rho_1^D = 1$. All leaves have $\rho_* t < R_w \leq t$, and thus

$$(6.5) \quad t^{-D} \leq N_R(t) \leq (\rho_* t)^{-D}.$$

The comparison proves the full-set limit γD .

By the area bound of Lemma 6.1, the type probabilities of actual assigned codes are bounded by C_R^P times their Bernoulli weights. The proof of Lemma 5.1 therefore gives one uniformly typical compact T_ζ of area greater than $1 - \zeta$. Set $K_\zeta = T_\zeta \cup \{0\}$ before choosing any exponent, sequence, or horizon. On this same set, for each $\eta > 0$ a constant $A_\eta \geq 1$ satisfies

$$(6.6) \quad \begin{aligned} A_\eta^{-1} e^{-(h+\eta)k} &\leq \lambda_{c_k(x)} \leq A_\eta e^{-(h-\eta)k}, \\ A_\eta^{-1} e^{-(\chi+\eta)k} &\leq R_{c_k(x)} \leq A_\eta e^{-(\chi-\eta)k}. \end{aligned}$$

Both follow from the same uniform frequency convergence. For any positive-area compact K , the same construction supplies a positive-area compact $T \subset K$ with these bounds.

Agreement with an arbitrary input through m has area at most $C_R^P A_\eta e^{-(h-\eta)m}$. A nonempty first-difference class at time $j \leq m$ has a common typical prefix. Equations (6.4) and (6.6) bound its area by a constant times $\delta e^{(\chi-h+2\eta)(j-1)}$. Since $\chi > h$, summation gives, uniformly in the input,

$$(6.7) \quad \text{area}(T \cap V_G(n, \delta, u)) \leq A'_\eta [e^{-(h-\eta)m} + \delta e^{(\chi-h+2\eta)m}].$$

For $m = \min\{n, \lfloor (-\log \delta)/(\chi+\eta) \rfloor\}$, the second term is at most $e^{-(h-\eta)m}$. Since $\gamma < \gamma_0 \leq \chi$, covering the positive area of T yields lower rate $\gamma h/\chi$, after $\eta \downarrow 0$.

For the matching upper rate on K_ζ , use all radial stopping prefixes realized on T_ζ , with the full tails from Lemma 6.2. Put $T = -\log \delta$ and $c_{\max} = -\log \rho_*$. Each such stop has length at most

$$\ell \leq \frac{T + \log C_{0,R} + c_{\max} + \log A_\eta}{\chi - \eta}.$$

At a fixed length ℓ , distinct realized words have $\lambda_w \geq A_\eta^{-1} e^{-(h+\eta)\ell}$, whereas all length- ℓ word weights sum to one. Their number is at most $A_\eta e^{(h+\eta)\ell}$. Summing up to the displayed length, and adding one input for the apex, gives upper rate $\gamma(h+\eta)/(\chi-\eta)$. Taking $\eta \downarrow 0$ proves the second limit in (2.11), on one fixed K_ζ for all the stated sequences.

Finally let $b_i = \rho_i^D$, so $b_0 + b_1 = 1$. Then

$$(6.8) \quad D\chi - h = \sum_i a_i \log(a_i/b_i) \geq 0.$$

The inequality follows from $-\log t \geq 1 - t$; equality holds only when $a_i = b_i$ for both digits. This would imply $\beta_0 D = \beta_1 D = 1$, which is impossible under mismatch. Hence $D > h/\chi$. The root and the strict entropy comparison are classical counting, used only after the physical catalogue comparison has been established. $\square$

Proposition 6.3 (Matched radius-autonomous subfamily). *If $\beta_0 = \beta_1 = \beta$, the system (2.9), on its full range (2.8), has all the rate conclusions of Theorem 2.2, including the single compact set for every finite high-side exponent.*

Proof. Here $R_w = \lambda_w^\beta$ and $\mathcal{R}_w = \mathcal{Q}_w^\beta$. Lemma 6.1 gives the full intervals and area bounds required in Lemma 5.1. Equation (6.4) becomes a one-sided physical band of area at most $C_{b,R} \delta \lambda_a^{1-\beta}$. Equation (6.3) becomes (4.4) with fixed constants and angular stopping thresholds $\delta^{1/\beta}$ and $(\delta/C_{0,R})^{1/\beta}$. Its upper count follows from (4.3); the 2^n exact bound also holds. Thus the proofs of Lemma 5.2 and Theorem 2.2 use exactly these already derived estimates: choose the same m for the lower bound, one uniformly typical compact before all exponents for the high-side upper bound, and the type t_γ in (5.6) for the full-set

strict lower bound. Every inequality and tail needed for those deductions has been supplied in this section. $\square$

Corollary 6.4 (Matching criterion within the radius-autonomous class). *For a fixed system (2.9), there is a nonempty interval of exponents adjacent to zero on which all fixed positive-area compact initial sets have the same rate for every corresponding precision sequence if and only if $\beta_0 = \beta_1$.*

Proof. Under equality, Proposition 6.3 gives the interval $0 < \gamma \leq \beta h$. Under inequality, Theorem 2.3 separates Q and one fixed nearly full-area compact set for every $0 < \gamma < \gamma_0$. $\square$

This criterion concerns only (2.9). It does not give a necessity statement for (2.5), overlapping feasible branches, or general smooth perturbations.

7. GENUINE FEEDBACK AND THE SCOPE OF COORDINATE REDUCTION

Proposition 7.1 (Nonnested feasible images). *For the feedback system (2.5), assume additionally*

$$(7.1) \quad 0 < |a_0 - a_1| < \varepsilon/4.$$

There is no homeomorphism preserving Q and simultaneously conjugating both controls to the matched linear maps or to any radius-autonomous system (2.9), even after permuting the input labels. The conjugacy identities are required on Q , with its domain containing $Q \cup H_0(Q) \cup H_1(Q)$. In particular, no fixed C^1 or bi-Lipschitz coordinate change of this kind reduces these feedback instances to those models.

Proof. Let $D_i = Q \cap H_i^{-1}Q$ and $J_i = H_i(D_i)$. For (2.9), $g_i(s) = sq_i(s)^{\beta_i}$ is increasing, independent of the angle, and the feasible angular image is all of $[-1, 1]$. Its feasible image is therefore the triangle

$$\{(s, z) : 0 \leq s \leq g_i(1), |z| \leq s\}.$$

The two images are nested. This also holds for the matched linear model.

For the feedback system put $\xi = a_0 - a_1$ and $\theta_1 = \theta(1)$. Equation (3.1) gives $|\theta_1| \leq |\xi|/(1 - \varepsilon)$. The largest radial coordinates on the lower and upper output rays of the two feasible images are

J_0	$(a_0 - \varepsilon/3)^\beta$	$((1 + \theta_1)/2)^\beta$
J_1	$((1 - \theta_1)/2)^\beta$	$(a_1 - \varepsilon/3)^\beta$.

To verify that these are maxima, not just sample values, note from (3.4) that each output ray has as preimage the relevant outer source ray or the dividing curve $y = \theta(s)$. On an outer ray the derivative of $sP_i(s, \pm 1)^\beta$ is positive because $p_- - \beta\varepsilon > 0$. On the divider, implicit differentiation gives $|\theta'| \leq 2\varepsilon/(1 - 2\varepsilon)$ and $|dP_i(s, \theta(s))/ds| \leq \varepsilon/(1 - 2\varepsilon)$. Thus the radial derivative is also positive there, since $p_- - \beta\varepsilon/(1 - 2\varepsilon) > 0$. Each maximum is attained at $s = 1$, with the full ray segment from zero included. At the divider $P_0 = (1 + \theta)/2$ and $P_1 = (1 - \theta)/2$; on the outer rays the feedback term at $s = 1$ is $\pm\varepsilon/3$. These facts establish the table.

Put $d = (\xi + \theta_1)/2$. Under (7.1), $|d| \leq |\xi|/(1 - \varepsilon) < \varepsilon/3$, since (2.3) implies $\varepsilon < 1/64$. The difference of the lower-ray bases is $d - \varepsilon/3 < 0$, whereas that of the upper-ray bases is $d + \varepsilon/3 > 0$. Hence J_1 extends farther on the lower ray and J_0 farther on the upper ray. Their images are not nested.

A simultaneous homeomorphism taking Q onto Q maps D_i and J_i to the corresponding target feasible domains and images. This follows from the conjugacy identities on Q and bijectivity; membership in Q is preserved in both directions. Such a map preserves inclusion, even with label permutation. It cannot change nonnested images into nested ones, proving the proposition. $\square$

The additional range is nonempty: take a_0, a_1 sufficiently close to $1/2$ and unequal, and choose a permitted ε with $4|a_0 - a_1| < \varepsilon$. Outside that range this argument does not decide coordinate reduction. It also does not exclude changes of coordinates that change the constraint set. Neither qualification restricts the boundary theorem on its full range.

8. LOCAL MATCHING DOES NOT SUFFICE

The following example keeps the common attracting origin and even agrees with the linear maps on a neighbourhood of it. It violates global injectivity by collapsing positive-area regions onto an invariant ray.

Proposition 8.1. *For every choice of a_0, a_1, β in (2.2) and every $0 < \sigma < 1/2$, there are C^∞ maps $S_0, S_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with these properties:*

- (1) *They equal A_0, A_1 on $\{s \leq \sigma\}$, so all derivatives at the origin agree.*
- (2) *All inputs drive bounded sets uniformly exponentially to the origin, and Q is controlled invariant.*
- (3) *The compact set $K_\sigma = Q \cap \{s \geq 2\sigma\}$ has area $1 - 4\sigma^2$ and $r_S(n, \delta; K_\sigma) = 1$ for every n and every $\delta > 0$.*
- (4) *For every finite $\gamma \geq 0$ and every sequence $\delta_n \rightarrow 0$ with $-\log \delta_n/n \rightarrow \gamma$, the full-set cost satisfies*

$$\min\{h, \gamma/\beta\} \leq \liminf \frac{\log r_S(n, \delta_n; Q)}{n} \leq \limsup \frac{\log r_S(n, \delta_n; Q)}{n} \leq \min\{\log 2, \gamma/\beta\}.$$

In particular, for $0 < \gamma \leq \beta h$ the full-set rate is γ/β , but the positive-area set K_σ has rate zero.

Proof. Let $e(t) = 0$ for $t \leq 0$ and $e(t) = \exp(-1/t)$ for $t > 0$, and define

$$\psi(s) = \frac{e(s - \sigma)}{e(s - \sigma) + e(2\sigma - s)}.$$

The denominator is positive everywhere. Thus ψ is smooth, zero for $s \leq \sigma$, and one for $s \geq 2\sigma$. Put $\rho_i = a_i^\beta$ and define

$$(8.1) \quad S_0(s, z) = (\rho_0 s, (1 - \psi(s))a_0^{\beta-1}(z + a_1 s) - \psi(s)\rho_0 s),$$

$$(8.2) \quad S_1(s, z) = (\rho_1 s, (1 - \psi(s))a_1^{\beta-1}(z - a_0 s) + \psi(s)\rho_1 s).$$

They have the claimed local identity. Each $S_i(s, z)$ is a convex combination of $A_i(s, z)$ and $R_i(s, z)$, where $R_0 = (\rho_0 s, -\rho_0 s)$ and $R_1 = (\rho_1 s, \rho_1 s)$. All four linear maps contract toward zero in one adapted norm $\max\{M|s|, |z|\}$ for sufficiently large M : the second-coordinate factor for A_i is at most $a_i^{\beta-1} + a_i^{\beta-1}a_{1-i}/M < 1$, and R_i has factor at most ρ_i . Convexity of the norm gives the same contraction bound for S_i on all of $\mathbb{R}^2$.

For controlled feasibility let $t = \psi(s)$, $y = z/s$, and $\xi = a_0 - a_1$. If $t < 1$ the full feasible intervals are

$$J_0(s) = \left[-1, \min \left\{ 1, \xi + \frac{2a_0 t}{1-t} \right\} \right],$$

$$J_1(s) = \left[\max \left\{ -1, \xi - \frac{2a_1 t}{1-t} \right\}, 1 \right].$$

They follow by imposing $-1 \leq z'/s' \leq 1$ in (8.1)–(8.2), and their union is $[-1, 1]$. For $t = 1$ both branches are feasible on the whole fibre. The radius always decreases, so Q is controlled invariant.

On K_σ , one application of S_0 sends every state to the lower ray. At every radius $S_0(s, -s) = (\rho_0 s, -\rho_0 s)$, so input 0^∞ keeps the whole set exactly in Q . Its area is $\int_{2\sigma}^1 2s ds = 1 - 4\sigma^2$.

To obtain the full-set lower bound, use $Q_c = Q \cap \{s \leq c\}$ with $c = \sigma/2$. For every input, the radius of a state in Q_c stays at most c . Thus $\psi = 0$ forever along that trajectory, even when its angle leaves $[-1, 1]$, and its entire trajectory is exactly that of A_i . Consequently $r_S(n, \delta; Q) \geq r_A(n, \delta; Q_c)$. Corollary 5.3 and (2.6) give the lower bound.

For the upper bound choose an integer t_0 such that $(\max_i \rho_i)^{t_0} \leq \sigma$. Every initial state admits an exact feasible prefix of length t_0 , ending in the region $s \leq \sigma$. All subsequent trajectories there coincide with the linear system, for all inputs, since the radius continues to decrease independently of the angle. Concatenate all 2^{t_0} possible prefixes with a catalogue for the linear full set over the remaining horizon. For $n \geq t_0$ this gives

$$r_S(n, \delta; Q) \leq 2^{t_0} r_A(n - t_0, \delta; Q) \leq 2^{t_0} C_U \delta^{-1/\beta}.$$

The 2^n exact-prefix bound also holds. These yield the upper rates for every stated precision sequence. Finally $\det DS_i = (1 - \psi(s))a_i^{2\beta-1}$ vanishes on the saturated region, identifying the noninjective mechanism. $\square$

For fixed σ this proposition gives one large zero-cost initial set in one fixed system. Making its area arbitrarily close to one changes σ and the system. This differs from Theorem 2.2, where all K_ζ are chosen inside a single fixed system. The collapse example is a counterexample to sufficiency of local matching and uniform convergence; it is not a counterexample to the reversible theorem.

9. SCOPE AND CONSEQUENCES

The common-cost boundary is a property of actual initial-set coverage, rather than convergence of controlled states alone. Below the boundary, a positive-area typical portion supplies a lower bound for every compact set, including sets with empty interior. The singular $1/s$ in the first-deviation band is cancelled by the physical area element. Above the boundary, rare terminal types force more controls for the full set, while one uniformly typical compact set excludes those persistent deviations. Its selection precedes all precision exponents.

Angular feedback removes the shared-radial-trajectory property of radius-autonomous words. The graph cone, complete feasible intervals, derivative comparison, and physical first-deviation band supply the missing connection to arbitrary approximate controls. The topological obstruction shows genuine instances of this extension within its stated subrange. The separate radial mismatch theorem makes precise how unequal matching powers destroy an initial common-cost interval. Its pressure root is meaningful for the real catalogue only after the relative-prefix spacing comparison has been proved.

The conclusions retain important restrictions. The feedback family has a common instantaneous power, nonoverlapping feasible branches, and a specific small perturbation with summable effect under origin contraction. We prove no general perturbation theorem from invertibility or a matching first derivative alone. The matching criterion belongs to the radius-autonomous class; necessity in an unequal-power angular-feedback class is not established here. The full-set high-precision bounds of Theorem 2.2 do not give an exact spectrum or its limit. These limitations concern the generality of the result, not gaps in the stated proofs.

Generative AI disclosure. Generative AI was used extensively in developing mathematical arguments, checking proof chains and related literature, drafting and restructuring the English text, and producing the LaTeX source. These activities do not constitute verification by a human author. Author identity, affiliations, authorship approval and the scope of human mathematical review remain to be supplied and confirmed before any submission.

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